MARISSA TONG

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SUMMARY

Software Engineer with 4+ years of experience building backend services, cloud-native systems, and data platforms at Solve and Amazon. Proven track record of cutting infrastructure and vendor costs while improving reliability. Skilled in building and shipping scalable backend services and customer-facing features using Ruby on Rails, SQL, and AWS.

EDUCATION

University of California, Irvine Irvine, CA
Bachelor of Science, Computer Science and Engineering

EXPERIENCE

Software Engineer II Jul 2026 – Present
Solve New York, NY (Remote)

- Made complex audience segmentation self-serve for non-technical users by designing and implementing a domain-specific language (DSL) engine that translates natural language queries into optimized SQL
- Reduced vendor costs by $2,000+/month and prevented projected $20,000/month spend by building an in-house replacement for Fivetran end to end with in-house OAuth integration
- Delivered the multi-tenant access-control layer for the company's data warehouse using AWS Lake Formation row-level security over Athena, Glue, and Apache Iceberg, replacing folder-based partitioning without changing customer delivery formats
- Automated visual preview generation for every published campaign across the platform by building and deploying a standalone rendering microservice using TypeScript, Express, Playwright, Docker, Terraform

Software Engineer Jul 2023 – Present
Solve New York, NY (Remote)

- Reduced infrastructure costs 40% and improved system performance by migrating legacy single-tenant customer stacks to a shared multi-tenant platform — Ruby on Rails, PostgreSQL, GraphQL, Sidekiq/Redis, Kubernetes, Terraform, and AWS (Kinesis, S3, Lambda, Athena, Glue)
- Drove a 5.7x increase in abandoned cart revenue and client engagement through integration of third-party marketing platforms (Cordial, Bluecore, Klaviyo) via REST APIs and OAuth
- Removed recurring manual reporting work for internal teams by automating report generation from external APIs into Google Sheets

Software Development Engineer Jul 2022 – Jul 2023
Amazon Web Services Seattle, WA (Remote)

- Reduced costs by 50% and improved operational efficiency by owning production systems supporting internal AWS-based services, built on ECS, S3, SNS, SQS, DynamoDB, and Kinesis
- Cut incident resolution time by implementing ECS Exec for live-container debugging, and supported high availability through on-call, monitoring, and incident response
- Increased release reliability and developer productivity by adding build-promotion workflows to deployment pipelines

RESEARCH EXPERIENCE

Undergraduate Research Assistant Sep 2020 – Oct 2022
UC Irvine Irvine, CA

- Built data pipelines to clean, transform, and geolocate IP-based datasets
- Maintained and optimized codebases for data analysis and reporting workflows
- Developed Python scripts to process and analyze large datasets, generating insights from 5+ years of student data

SKILLS

Languages: Java, Python, Ruby, SQL, C++
Backend & Cloud: Claude, Ruby on Rails, REST & GraphQL API design, AWS (Lambda, ECS, S3, SQS, SNS, Kinesis, Athena, Glue, DynamoDB), Kubernetes, Terraform, Sidekiq, Redis, CI/CD
Data & Systems: PostgreSQL, data warehousing, Apache Iceberg, distributed systems, event-driven architecture, performance optimization